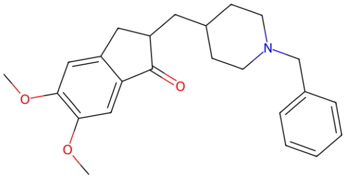
A the molecule: donepezil

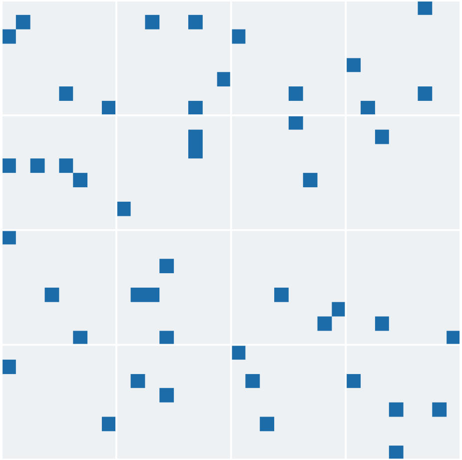


`COc1cc2c(cc1OC)C(=O)C(CC1CCN(Cc3ccccc3)CC1)C2`

standardised first: largest organic fragment, salts stripped, sanitised, then keyed by the InChIKey of that parent

B the fingerprint: 1024 bits, 47 set

each cell is one bit, read left to right, top to bottom



Morgan / ECFP-4, radius 2, folded to 1024 bits, chirality NOT included. Folding means a set bit reports that some environment hashing to that index is present, not which one: several substructures share a bit. Excluding chirality means two enantiomers produce identical rows, which is why identical rows are collapsed before any split rather than left to fall on both sides of one.

C the descriptors: 12 values

molecular weight	379.50
Crippen logP	4.36
TPSA (A^2)	38.77
H-bond donors	0.00
H-bond acceptors	4.00
rotatable bonds	6.00
aromatic rings	2.00
fraction sp3 C	0.46
rings	4.00
heavy atoms	28.00
formal charge	0.00
QED drug-likeness	0.75

Unscaled. A random forest splits on thresholds and is unchanged by any monotone rescaling, so no scaler is fitted and none can leak across a split.

1024 + 12 = 1,036 columns, every endpoint